

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 415 — Past Questions & Materials
by Renance

Course: PEE 415
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PEE 415 Handout

Level 400 • First Semester • Handout

Faculty of Engineering

Petroleum Engineering Department

Oil properties Laboratory, 2nd stage

Experiment No. 5: Water content by Centrifuge

Prepared by: Brosk Frya Ali

10/11/2013

11/17/2013

Introduction

This test method describes the laboratory determination of water sediments in crude oil by means of centrifuge procedure. This centrifuge method for determining water sediments in crude oil is not entirely satisfactory. The amount of water detected is almost always lower than the actual water content. There is another method to determine the water content in crude oil called "Dean stark" method and it consider the best method.

11/17/2013

Figure 1: General appearance.

11/17/2013

Aim of the experiment

- Determines the 'water content' in a sample of crude oil by the centrifuge device.

11/17/2013

Apparatus

- 1- Centrifuge force instrument.
- 2- Rubber Cushion (16203 & 16104).
- 3- Bucket (13113 & 13114).
- 4- Glass tube (100 ml).
- 5- Cork.

11/17/2013

Experimental Procedures:

GOGGLES may be required!

1. Fill two centrifuge tubes to the 50 ml with homogeneous sample to be tested. Read the top of the meniscus (the curved upper surface of a liquid column).
2. Fill the two tubes with 50 ml of solvent, which brings the total

contents of the tubes to the 100 ml the acceptable solvents are Stoddard, kerosene, toluene & xylene. Note that toluene & xylene should be water-standard at test temperature.

3. Invert the tubes 10 times to mix the sample & solvent. This should be done below eye level for safety, and safety glasses are highly recommended. Solvents can be hazardous and should be handled caution.

11/17/2013

4- place the tubes in the pre-heater and heat to 49C.

5- Invert the tubes 10 times again to mix.

6- place the tubes on the opposite sides of the centrifuge to balance the load. Close the centrifuge lid.

For a minimum of 10 min. At the speed [that can be calculated from equation (1)]. Enough to generate the relative force centrifuge (RCF) reach to 500 – 800 rpm.

7- Record the value of water & accumulation precipitates in the tube.

8- Return the tube to the centrifuge without shaking for 10 min. At the same speed to obtain the correct value.

9- The summation of water content & precipitate in each tube represent the percentage of water & precipitate.

11/17/2013

Calculations

11/17/2013

Discussion Section

1. Discuss the reasons of any error happened during the Exp. If applicable?

2. What is the effect of Increasing the RCF.?

3. What is the advantage of the placing two tubes at opposite side?

4. Discuss the disadvantage of water in crude oil.

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Paper 2 of 2 — PEE 415 Handout

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Petroleum Engineering Department

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